

# Md Khalequzzaman Sarker Likhon

Dhaka, Bangladesh · khalequzzamanlikhon@gmail.com · +880 1711 906648  
khalequzzamanlikhon.github.io · github.com/khalequzzamanlikhon · linkedin.com/in/khalequzzaman-likhon ·  
ORCID

## RESEARCH INTERESTS

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Understanding human and team behavior from multimodal data · Video LLMs · Grounded, verifiable AI explanations · Human-centered evaluation and measurement validity

## EDUCATION

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**Ahsanullah University of Science and Technology**

Apr 2016 – Jan 2021

*B.Sc. in Computer Science and Engineering*

Dhaka, Bangladesh

- **Thesis:** *Design of an Arrhythmia Classification Algorithm Using 2-D Convolutional Neural Networks* (Supervisor: **[Dr. Firstname Lastname]**)

## RESEARCH AND ENGINEERING EXPERIENCE

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**Machine Learning Engineer, Accelx Inc.**

Jan 2025 – Present

*Human behavior in video, vision-language models, and grounded LLM outputs for human reviewers* Dhaka, Bangladesh

### **Appearance or body dynamics for recognizing violent interaction**

*Which carries the signal for recognizing a fight in video: spatiotemporal appearance, or how people's bodies move?*

- On RWF-2000, a Kinetics-pretrained R(2+1)D-18 over 2s clips reaches 84.6% clip accuracy; a Transformer over YOLO11-Pose dynamics for up to 4 people (joint velocities, arm angles, an inter-person contact cue) reaches 61.7%, and a pose-RGB fusion model 69.0%. Checkpoints were selected on the same split they are reported on, so these are optimistic.
- Audited the pose features rather than the score: ~70% of the “two people in contact” signals came from joints missing for *both* people, not from real proximity, and person slots are not tracked between frames, so the model can report that a fight happened but not who did what.

Deployed video models for fall and violence detection in live multi-camera monitoring at sub-100 ms latency. **[number of cameras or sites]**

### **Vision-language models under deployment shift**

*How far can a frozen CLIP model go on rare events without labeled video, and how does it fail on real footage?*

- Built zero-shot fall and violence recognizers on OpenCLIP ViT-L/14 with prompt banks and an F1-calibrated decision rule. Error analysis on real footage traced false positives to overhead and fisheye views, shadows, seated postures and hugging; mined 80+ hard-negative prompts from them as a training-free correction. **[false-positive rate before → after]**

### **Grounded LLM and VLM outputs that support human reviewers**

*How can small local models produce explanations a human can verify?*

- A local LLM writes only from facts computed in code; every citation is checked by a word-overlap test whose threshold was set from observed failures (~17% overlap for misattributed citations vs. ~100% for correct ones).
- A staged image-analysis cascade (OCR, CLIP, then Qwen3-VL only for hard cases) verifies findings against a knowledge base and abstains to human review when evidence is weak. **[share of images sent to the VLM; accuracy]**

**[Role, Organization]**

**[One honest line: what you did]**

**[2021 – 2024]**

**[City]**

## RESEARCH PROJECTS

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### Telemetry vs. video for automated after-action review [GitHub]

Independent pilot, Sep 2026

*When the same model writes an after-action review from a game log and from gameplay video, which of its claims hold up?* On 15 human–human Overcooked episodes (Carroll et al., 2019), Qwen2.5-VL reads each source one minute at a time and returns timestamped claims, each checked against the log (1,140 claims, 45 windows). Log claims were supported 81% of the time vs 53% from video, and the gap is timing, not invention: 98% of video timestamps fell on a 5 s grid (vs 1%), and video under-counted deliveries in 41 of 45 minutes. Blind pairwise human rating in progress.

### DocuRoute: evaluating retrieval choices for grounded question answering [GitHub]

2026

LLM routing between hybrid retrieval and sandboxed text-to-SQL over SEC 10-K filings, with cited answers and a custom evaluation harness. A general-domain cross-encoder reranker lowered faithfulness from 0.93 to 0.77, suggesting sensitivity to domain shift.

**Other engineering projects:** Sentinel, tracking-based video anomaly rules at 19.7FPS [GitHub] · SentinelRAG [GitHub] · Research→Code [GitHub] · more at [khalequzzamanlikhon.github.io/projects](https://khalequzzamanlikhon.github.io/projects)

## MANUSCRIPTS

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\* equal contribution

- **M. K. S. Likhon\***, M. M. Sarker\*. *Out-of-Focus Is Not Obliteration: A Perturbation Analysis of Fingerprint Alteration Detectors on Synthetic Benchmarks*. Manuscript in preparation, 2026.  
Detectors reaching 0.998 macro-F1 on SOCOFing respond to editing artifacts rather than to the alteration they are meant to detect.
- **M. K. S. Likhon\***, M. M. Sarker\*. *Does Geometry Help? A Leakage-Audited, Negative-Result Study of Cross-Attention Fusion and Taxonomy-Consistency Loss for Post-Earthquake Damage-Type Classification*. Manuscript in preparation, 2026.  
Geometry fusion performs no better than a shuffled-geometry control on PEER Hub ImageNet.
- R. Tanvir\*, **M. K. S. Likhon\***, **[all remaining authors, in order]**. *Joint Prediction of Ultra-High-Performance Concrete Properties Using a Feature-Tokenizer Transformer with Masked Multi-Task Learning*. Under review at *Applied AI Letters* (Wiley), 2026.

## TECHNICAL SKILLS

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|             |                                                                                                                                                                                          |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods     | Video LLMs and VLMs (Qwen2.5-VL, Qwen3-VL, CLIP); action recognition; pose estimation and tracking; transformer fine-tuning; LLM grounding and RAG; evaluation design and error analysis |
| Frameworks  | PyTorch, Hugging Face Transformers, OpenCLIP, Ultralytics, OpenCV, scikit-learn, W&B                                                                                                     |
| Engineering | Python, C/C++, SQL, Bash; FastAPI, Docker, TorchServe, Redis, Git, Linux                                                                                                                 |

## TEST SCORES

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**GRE:** Quantitative 155, Verbal 151, Analytical Writing 3.0 · **IELTS:** 7.0 (Listening 8.0, Reading 6.5, Writing 6.5, Speaking 6.5)

## TEACHING AND SERVICE

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| Home tutor for secondary and higher-secondary students (grades 10–12) | 2017 – 2022 |
| Organizing Committee, AUST Codeware (programming contest)             | 2019        |

## REFERENCES

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Available on request. **[ask three referees before you apply; Memphis requires three letters]**